

Recession Forecasting using the Yield Curve under Regime Change

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Abstract

This paper examines the predictive power of the yield curve for US recessions under regime change, focusing on the Global Financial Crisis (2007-2009) and COVID-19 pandemic (2019-2021). We decompose the yield curve into Level, Slope, and Curvature factors using Principal Component Analysis and demonstrate fundamental regime dependence in their forecasting performance. Using machine learning methods including Logistic Regression, Ridge, Elastic Net, Random Forests, and Gradient Boosting, we show that yield curve factors successfully predict financially-driven recessions (0.833 balanced accuracy during GFC) but fail during exogenous shocks, where macro indicators dominate (0.857 balanced accuracy during COVID-19). We adjust for term premium effects using Kim-Wright estimates and document substantial factor loading drift during unconventional monetary policy periods. Rolling PCA analysis reveals structural instability in long-maturity yields around quantitative easing episodes. Our findings indicate that robust recession forecasting requires adaptive ensemble approaches that dynamically weight financial signals during monetary tightening cycles and real-sector indicators during external disruptions.

1 Introduction

Economic recessions represent periods of significant contraction in economic activity that impose substantial costs on society through job losses, reduced output, and diminished welfare. Accurately forecasting these downturns remains a central challenge in macroeconomics and financial economics, with important implications for policymakers, investors, and the broader public (Murphy et al., 1959). Among the various indicators examined in the literature, the yield curve, the term structure of interest rates across different maturities, has emerged as one of the most reliable predictors of future economic contractions.

The predictive power of the yield curve, particularly the spread between long-term and short-term Treasury yields, has been well documented since the seminal work of Estrella and Mishkin (1996). Their probit-based framework demonstrated that inversions of the yield curve, where short-term rates exceed long-term rates, reliably signal recessions four to six quarters ahead. The economic intuition underlying this relationship is compelling: an inverted curve reflects market expectations of future monetary policy easing in response to anticipated economic weakness, as investors price in lower future short-term rates relative to current levels. Recent work by Johansson and Meldrum (2018) and Sharpe and Engstrom (2018) has refined this approach by focusing on near-term forward spreads, which more directly capture market expectations of monetary policy trajectories over the next 12-18 months.

However, the post-2008 financial landscape has been characterized by unprecedented monetary policy interventions, including quantitative easing programs, forward guidance, and extended periods at the zero lower bound. These unconventional policies have fundamentally altered the behavior of long-term yields by compressing term premia, the additional compensation investors demand for holding longer-maturity bonds. As Rudebusch et al. (2006) emphasize, movements in long-term rates embody both changes in expectations about future short rates and variations in the term premium, and these two components can have markedly different macroeconomic implications. During periods of aggressive asset purchases by central banks, term premium compression can flatten or even invert the yield curve without necessarily signaling expectations of future rate cuts, thereby potentially generating false recession signals. Bauer et al. (2018) argue that

adjusting for term premia does not necessarily improve forecast accuracy in model-free settings, though they acknowledge the importance of distinguishing these components for structural interpretation.

Moreover, the COVID-19 pandemic introduced an entirely new type of economic shock, an exogenous public health crisis that triggered sudden stops in economic activity largely independent of financial market conditions. This episode raised fundamental questions about whether traditional financial indicators, including yield curve factors, retain their predictive power during non-financial crises. The sharp divergence between financial signals and real economic outcomes during 2020 suggests that recession forecasting models must be adaptive to different crisis regimes.

This paper makes several contributions to the literature on yield curve-based recession forecasting. First, we extend the traditional spread-based approach by decomposing the entire yield curve into three latent factors, Level, Slope, and Curvature, using Principal Component Analysis rather than relying solely on parametric specifications such as the Nelson and Siegel (1987) framework or the dynamic extension proposed by Diebold and Li (2006). This data-driven approach captures the full dimensionality of term structure movements and allows us to examine how each factor’s relationship with future recessions varies across regimes. Second, we explicitly address the term premium problem identified by Rudebusch et al. (2006) by incorporating Kim-Wright decompositions to construct adjusted slope measures that isolate expectations-driven yield movements from non-expectational forces. Third, we move beyond linear probit models to employ modern machine learning techniques, including Ridge Regression, Elastic Net, Random Forests, and Gradient Boosting, that can flexibly capture nonlinear relationships and structural breaks. Fourth, we implement a rigorous out-of-sample evaluation framework with strict time-respecting procedures to avoid look-ahead bias and assess true real-time forecasting performance.

Our empirical analysis spans from 1982 to 2025 and includes detailed scenario studies of the two most recent major recessions: the Global Financial Crisis and the COVID-19 recession. We construct multiple feature sets combining yield curve factors with real-sector macroeconomic indicators such as unemployment, industrial production, and inflation to

test whether hybrid models can adapt across different shock types. The results reveal striking regime dependence: during the GFC, yield PCA factors dominate with balanced accuracy exceeding 0.83, while during COVID-19, macro-only models achieve 0.86 balanced accuracy as yield signals fail almost entirely. Feature importance analysis confirms that the slope factor (PC2) and traditional spreads are most informative during financial crises, whereas the level (PC1) and curvature (PC3) factors gain prominence during periods of intense policy intervention.

These findings have important practical implications. For policymakers and market participants, our results suggest that single-indicator models are insufficient in the post-crisis era. Instead, robust recession forecasting requires ensemble approaches that dynamically weight financial versus real-sector signals based on the prevailing economic regime. We conclude by discussing potential extensions, including regime-switching meta-models, integration of high-frequency data such as weekly jobless claims, and further decomposition of yield factors into risk-neutral and term premium components.

2 Literature Review

In macro and financial economic forecasting, the yield curve has been documented as the leading indicator for economic recessions. This literature review examines the theoretical and empirical foundations of yield curve-based recession forecasting, with particular emphasis on factor decomposition methods, term premium adjustments, and the evolution of predictive modeling techniques.

Estrella and Mishkin (1996) establish the foundational empirical framework for using the yield curve to predict U.S. recessions. Using data from 1960 to 1995, they demonstrate that the spread between the 10-year Treasury note and the 3-month Treasury bill is a powerful predictor of recessions at horizons of 2 to 6 quarters ahead. Their probit model framework shows that the yield spread significantly outperforms other widely used forecasting measures, including the stock market, the Commerce Department's index of leading economic indicators, and various monetary aggregates. The authors find that a spread greater than 1.2 percentage points corresponds to a recession probability of

less than 5 percent, while the probability rises to approximately 25 percent when the spread narrows to zero, and jumps to 50 percent when the curve inverts to a negative 0.8 percentage point spread. The predictive strength stems from the yield curve's role as a reflection of market anticipations regarding future monetary policy easing. When investors expect economic contraction, they anticipate that the Federal Reserve will lower short-term rates, causing the curve to flatten or invert. The paper's out-of-sample forecasting tests demonstrate robust predictive performance across different time periods, establishing the yield spread as one of the most reliable leading indicators available to policymakers and market participants.

Diebold and Li (2006) propose a dynamic Nelson-Siegel framework that revolutionizes yield curve modeling by reducing the entire term structure to three interpretable latent factors: level, slope, and curvature. Building on the classic Nelson-Siegel (1987) exponential components framework, they treat these three factors as time-varying parameters that evolve according to autoregressive processes. The level factor (β_1) represents long-term interest rates and shifts the entire curve vertically, exhibiting the highest persistence with correlation of 0.99 to empirical measures. The slope factor (β_2), defined as the negative spread between short-term and long-term rates, shows correlation of 0.99 with the traditional 3-month to 10-year spread and is closely connected to the monetary policy instrument. The curvature factor (β_3) captures the convexity or "hump" in the yield curve, with peak loading at medium-term maturities around 30 months. While their primary focus is on optimizing the accuracy of yield level forecasts rather than recession prediction, their framework proves particularly valuable for forecasting at horizons of 12 months and beyond, where it substantially outperforms random walk and Fama-Bliss forward rate regression models. The model's 12-month-ahead forecasts achieve significant improvements in root mean squared error at all maturities, often by wide margins. This data-driven factor extraction method provides a parsimonious yet highly effective representation of yield curve dynamics, capturing both cross-sectional fit and time-series evolution without imposing the strict no-arbitrage constraints of affine term structure models. The framework has become widely adopted by central banks globally for fitting and forecasting government bond yields.

Rudebusch et al. (2006) provide crucial insights into the dual nature of long-term interest rates by decomposing movements into two distinct components: changes in expectations about future short-term rates and variations in the term premium (the compensation investors demand for holding long-term bonds rather than rolling over short-term securities). Their analysis demonstrates that the macroeconomic implications of these two components can differ significantly, challenging traditional yield curve analysis that treats the spread as a monolithic indicator. Using data from the U.S. term structure and employing various estimation methods including the Kim-Wright model, they show that both monetary policy shocks and technology shocks affect the term premium and output in opposite directions—monetary and technology shocks raise the term premium but decrease output, while demand shocks increase both. The paper documents a significant decline in GDP volatility since 1984 (following McConnell and Perez-Quiros 2000) and an additional sharp drop between 2001 and 2005, suggesting that reduced macroeconomic volatility has contributed to lower term premia. They estimate that reduced volatility and increased Federal Reserve predictability may account for as much as 40 basis points of the decline in long-term relative to short-term yields during the mid-2000s monetary tightening cycle. This is particularly relevant for understanding the atypical 2004-2005 episode when the Federal Reserve increased the federal funds rate from 1 percent to 4.25 percent, yet 10-year Treasury yields fell from 4.7 percent to 4.5 percent—contrary to the typical positive correlation between short-term and long-term rates during tightening cycles. The research highlights that failing to account for term premium dynamics can lead to misinterpretation of yield curve signals, especially during periods of unconventional monetary policy such as quantitative easing or when interest rates approach the zero lower bound.

Bauer et al. (2018) provide a comprehensive empirical assessment of which term spread measures are most reliable for forecasting U.S. recessions and whether adjustments for term premia enhance predictive accuracy. Using monthly data from January 1972 to July 2018 and employing area under the curve (AUC) metrics to measure forecast accuracy, they systematically compare the predictive power of various spread measures for identifying recessions 12 months ahead. Their key finding is that the 10-year minus 3-month Treasury

spread emerges as the most reliable predictor, achieving an AUC of approximately 0.88, substantially better than the 0.5 that would correspond to a coin toss. This spread outperforms the more commonly cited 10-year minus 2-year spread ($\text{AUC} \approx 0.82$) and the 10-year minus 1-year spread. Importantly, they find that adjusting the term spread for estimates of the term premium—contrary to some theoretical arguments—does not improve recession forecasting accuracy and may even diminish it. Specifically, when they decompose the term spread into its expectations component and risk premium component using various term premium estimates (including Kim-Wright and Adrian-Crump-Moench models), both components carry similar signals about future recessions, but combining them or using adjusted spreads provides no additional forecast improvement beyond using the unadjusted spread alone. The authors argue that the yield curve requires an actual inversion to reliably signal recession—a merely flat curve is insufficient. Their analysis reveals that every U.S. recession in the preceding 60 years was preceded by a negative term spread, and except for one false positive in the mid-1960s, a negative term spread was always followed by an economic slowdown. They caution, however, that while these empirical correlations are robust, they do not establish causation, leaving open whether yield curve inversions cause recessions (perhaps through credit channel effects as banks face compressed net interest margins) or merely reflect market participants' recession expectations formed on other grounds. This distinction has important implications for policy interpretation during periods of yield curve flattening.

Building upon this established literature, our approach extends the foundational work of Estrella and Mishkin (1996) in several key directions. First, reflecting the market's collective conviction on future economic growth and monetary policy cycles, we adopt the factor decomposition methodology inspired by Diebold and Li (2006), but extract level, slope, and curvature factors through Principal Component Analysis rather than parametric Nelson-Siegel estimation, providing a purely data-driven representation unconstrained by functional form assumptions. Within this framework, the level factor (PC1_Level) represents long-term interest rates and inflation expectations, effectively shifting the entire curve vertically. The slope factor (PC2_Slope), defined as the spread between long-term and short-term rates, serves as the primary recessionary barometer; a negative slope

(inversion) implies that investors anticipate future rate cuts due to economic contraction. Finally, the curvature factor (PC3_Curvature) captures the "hump" or convexity of the curve, representing the movement of medium-term yields relative to the short and long ends, which is particularly sensitive to market volatility and unconventional policy interventions.

Following the insights of Rudebusch et al. (2006), who highlight that movements in long-term rates represent both changes in expectations about future short rates and variations in the term premium with potentially divergent macroeconomic implications, we incorporate term premium adjustments into our framework. Using the Kim-Wright 10-year term premium estimate (tp_10y_kw), we derive an adjusted slope series (slope_ex_tp) to discern whether curve inversion is driven by true expectations of monetary easing or a mere compression of the term premium—a distinction that is especially relevant during periods of Quantitative Easing (QE) or the Zero Lower Bound (ZLB). While Bauer et al. (2018) argue that adjusting for term premia does not necessarily improve accuracy in a model-free setting, we embed the term premium and adjusted slope directly into a multivariate framework that includes real-sector indicators, such as unemployment (unrate) and industrial production (indpro_yoy), to account for sudden exogenous shocks like the COVID-19 pandemic.

This theoretical decomposition is clearly visible in the transition from the pre-crisis to the recovery era shown in Figure 1. The 2006 average (blue line) exhibits a "high Level, zero Slope" dynamic, where short-term rates ($y_{3m} \approx 5\%$) have converged with long-term rates ($y_{30y} \approx 5\%$). The absence of Slope and Curvature here indicates a restrictive monetary environment where the term premium has eroded, a classic precursor to the 2008 recession. Conversely, the 2010 average (red line) illustrates a fundamental regime shift. While the level at the long end remains elevated, the slope has steepened dramatically ($S_t \gg 0$) as short-term rates approach the zero lower bound. Furthermore, the 2010 curve displays significant curvature (concavity), reflecting the market's nuanced pricing of medium-term recovery risk against a backdrop of aggressive monetary stimulus.

To address the limitations of linear probit models, our approach employs machine learning techniques such as Random Forests and Gradient Boosting, which allow us to capture regime dependency, structural changes, and complex nonlinear interactions

that may vary across different economic environments. Finally, we implement a strictly time-respecting, out-of-sample evaluation framework, ensuring that model parameters, classification thresholds, and feature selection are all determined without look-ahead bias, thereby providing a more robust real-time prediction of recessionary risks that would be operationally feasible for policymakers and market participants.

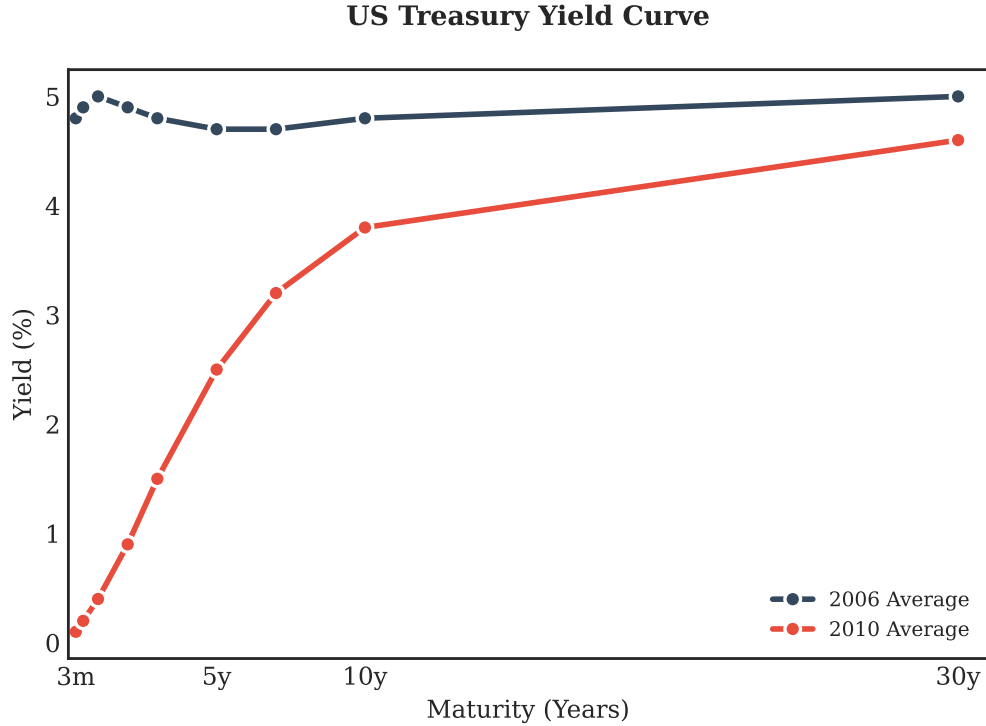


Figure 1: US Yield Curve for selected Maturities for 2006 and 2010

Source: FRED

3 Data

We focus our analysis on the United States over the period from 1982 until 2025 and adjust the timeframe analyzed based on the desired scenarios.¹ Our target variable are monthly recession indicators for the United States, as defined by the *National Bureau of Economic Research* (NBER). For the empirical replication of the yield curve, we use Treasury yields and macro indicators provided by the *Saint Louis Fed* (FRED). To ensure

¹Scenarios are the Great Financial Crisis in 2007-2009 and Covid-19 in 2019-2021.

a true forecasting horizon, the target is shifted backward by twelve months, such that all predictors available at time t are used to forecast recession status one year ahead at time $t + 12$. The yield curve dataset consists of monthly Treasury yields spanning different maturities from three months to thirty years, as released by the US Federal Reserve. These yields are used both directly and as inputs to principal component analysis.

Macroeconomic indicators include the unemployment rate, year-over-year CPI inflation, year-over-year industrial production growth, and consumer sentiment.² To distinguish expectations-driven yield curve movements from term premium effects, the ten-year yield is decomposed into an expectations proxy and a term premium component. An adjusted slope measure is constructed by subtracting the term premium from the long rate prior to computing the slope relative to the two-year yield. This allows direct testing of whether recession signals arise from anticipated policy easing or from non-expectational forces such as quantitative easing and safe-asset demand.

4 Principal Component Analysis of the Yield Curve

In order to obtain estimates for the latent factors of the yield curve, we chose to apply principal component analysis (PCA), because it effectively distills the high-dimensional, highly correlated movements of the term structure into a set of independent latent factors. By decomposing the covariance matrix of yields, PCA recognizes that the curve behaves like a smooth, continuous function rather than a set of isolated data points. The mathematical requirement for orthogonality then forces these factors into the "Level, Slope, and Curvature" hierarchy.

4.1 Explained Variance

Figure 2 presents the scree plot of the PCA applied to standardized yields across maturities. As expected, the first principal component explains approximately 97% of total variance, while the second and third components explain roughly 2.5% and 0.1%, respectively. This indicates that the U.S. yield curve is overwhelmingly driven by a common level factor,

²Consumer Sentiment as measured by the University of Michigan Survey

although slope and curvature components remain economically meaningful despite their small variance shares.

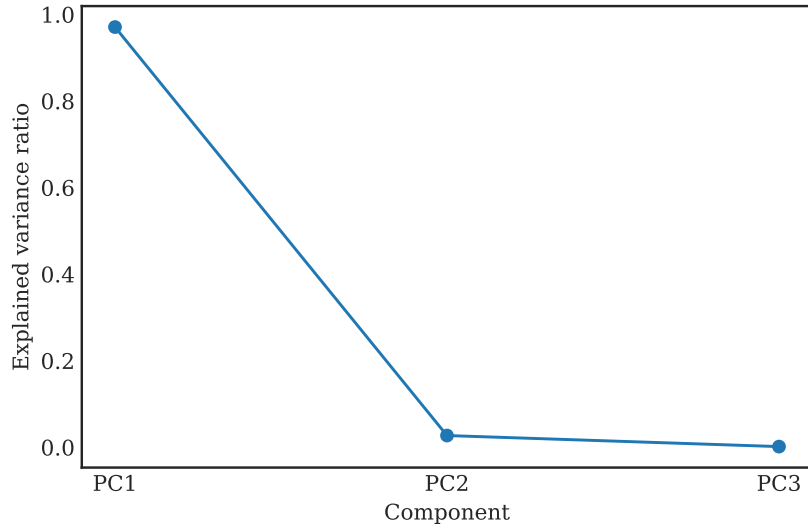


Figure 2: Explained variance ratio of yield curve principal components.

4.2 Factor Loadings

Got get a better picture of the shape of the curve, we look at the factor loadings at all maturities used. Figure 3 displays the PCA loading heatmap across maturities. The first principal component loads positively and nearly uniformly across all maturities, consistent with a level factor. The second component exhibits negative loadings at the short end and positive loadings at the long end, representing yield curve slope. The third component captures curvature, reflecting differential movements of the belly of the curve relative to short and long maturities.

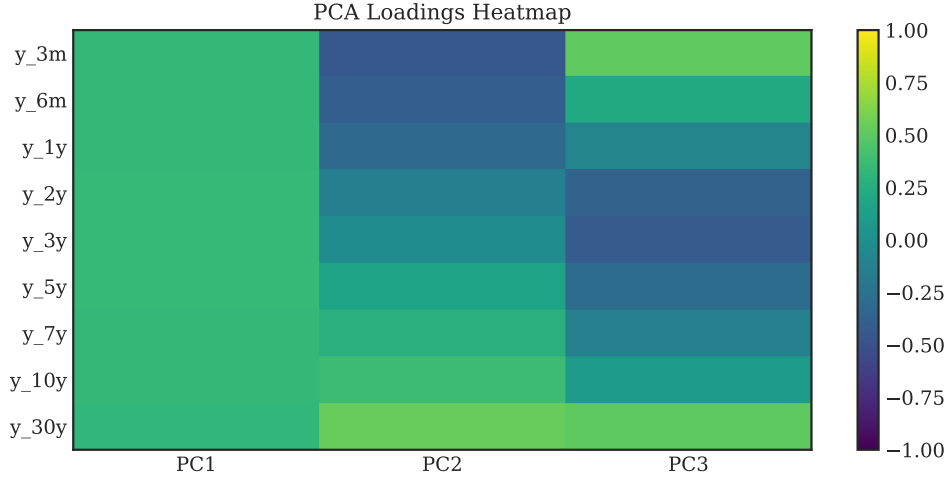


Figure 3: Heatmap of yield curve PCA loadings by maturity from 3 months until 30 years.

4.3 Factor Dynamics and Recessions

Figure 4 plots the time series of the first three principal components with NBER recession shading. The level factor reflects long-run secular declines in interest rates and major monetary regime shifts. The slope factor exhibits pronounced swings around tightening cycles and yield curve inversions preceding recessions. The curvature factor shows episodic spikes during periods of financial stress and policy intervention.

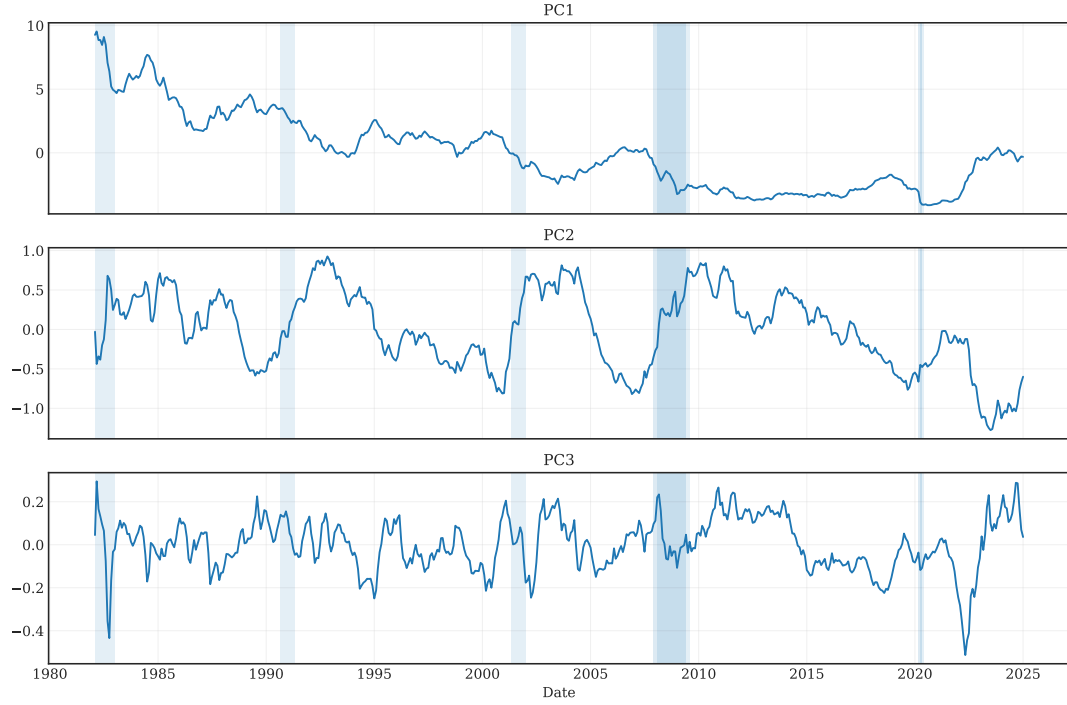


Figure 4: Time series of yield curve principal components with recession shading.

4.4 Rolling Loading Stability

Figures 5–7 present rolling-window PCA loading estimates for the first three components. While the overall factor structure is preserved, substantial drift is observed, particularly at the long end of the curve during the Global Financial Crisis and the COVID-19 period. These findings suggest that yield curve factors are regime-dependent and that long-maturity yields become structurally distorted during periods of unconventional monetary policy.

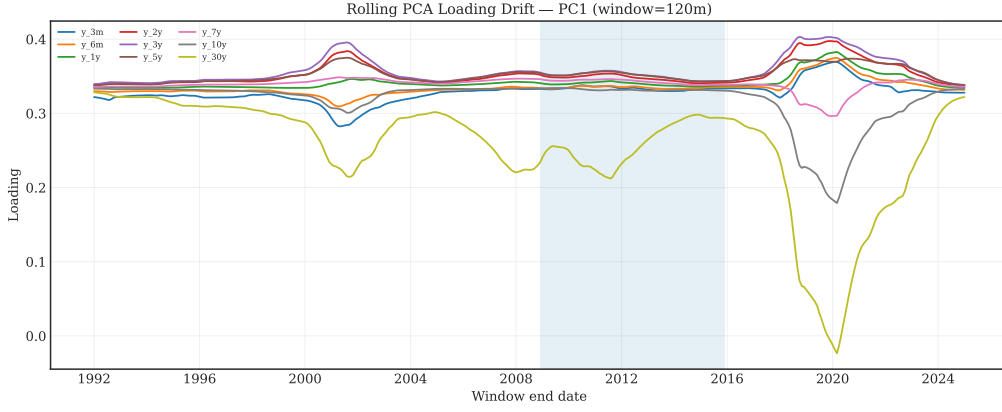


Figure 5: Rolling PCA loading drift for the level factor (PC1).

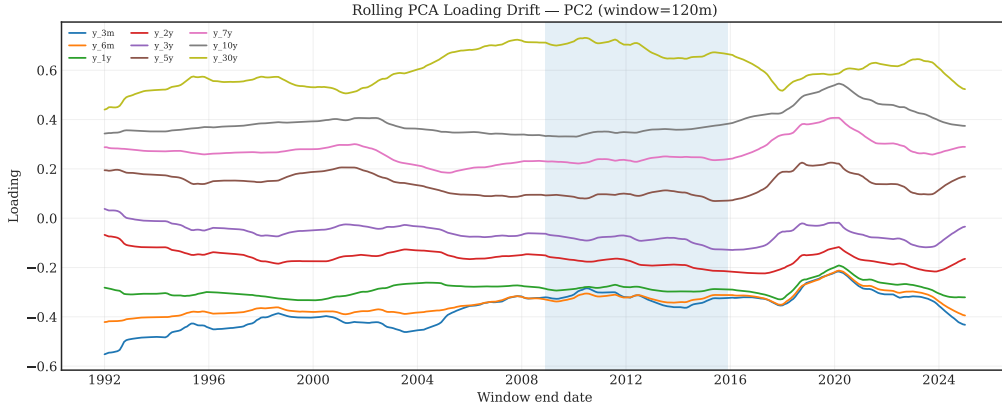


Figure 6: Rolling PCA loading drift for the slope factor (PC2).

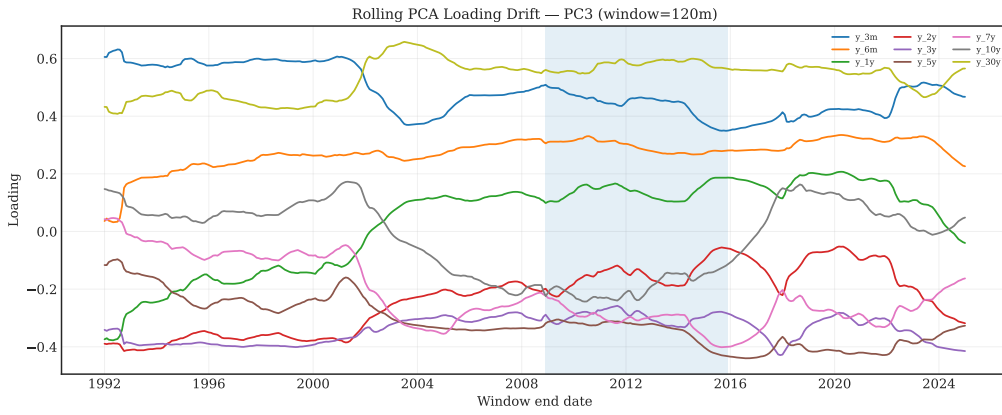


Figure 7: Rolling PCA loading drift for the curvature factor (PC3).

5 Recession Prediction

5.1 Models

To analyze the relationship the yield curve with recessions, we employ a broad set of machine learning estimators. As a baseline estimator we use a simple Logistic regression.

5.1.1 Regularization Approaches

We employ regularization-driven estimation to identify the optimal linear predictors while mitigating the risk of overfitting through the application of L_1 and L_2 penalty terms, commonly referred to as selection and shrinkage, respectively. Specifically, we utilize a Ridge estimator, which relies on L_2 regularization to perform coefficient shrinkage, and an Elastic Net estimator, which incorporates a weighted combination of both L_1 and L_2 penalties. The Ridge regression estimator is defined by the following optimization objective, which introduces the L_2 penalty to the residual sum of squares:

$$\hat{\beta}_{Ridge} = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\} \quad (1)$$

The Elastic Net estimator bridges the gap between Ridge and Lasso by incorporating a mixing parameter $\alpha \in [0, 1]$, which balances the L_1 and L_2 penalties:

$$\hat{\beta}_{EN} = \underset{\beta}{\operatorname{argmin}} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1 - \alpha}{2} \|\beta\|_2^2 \right) \right\} \quad (2)$$

To adapt these methods for classification, the optimization objective replaces the squared error used in regression with a log-likelihood loss function (log-loss) while maintaining the L_1 and L_2 penalty structures to constrain the model's coefficients.

$$\min_w \underbrace{\sum_{i=1}^n \log(1 + e^{-y_i x_i^T w})}_{\text{Classification Loss}} + \lambda \underbrace{\left[\rho \|w\|_1 + \frac{1 - \rho}{2} \|w\|_2^2 \right]}_{\text{Regularization Penalty}} \quad (3)$$

Given that our feature set is constructed based on a priori economic rationale and supported by established academic literature, we hypothesize that automated feature selection may provide limited utility in this context. This assumption would be empirically supported if the Ridge estimator demonstrates predictive performance comparable or superior to that of the Elastic Net.

5.1.2 Random Forests

We include tree-based methods to strengthen our approach by relying on their non-parametric estimation properties. At their core, these methods estimate the conditional expectation $E[Y|X]$ by recursively partitioning the feature space into disjoint regions and assigning a constant value to each. For regression tasks, the model estimates the conditional average of the target variable within each leaf node, while in classification, the estimate represents the conditional probability of a class, which is effectively the conditional average of the class indicator variables. While a single tree approximates these averages through a simple local mean, ensemble architectures like Random Forest and Gradient Boosting enhance this estimation by either averaging across multiple independent partitions to reduce variance or sequentially refining the conditional mean to minimize a global loss function.

While both Random Forest and Boosting are ensemble methods based on decision trees, they differ fundamentally in how they combine learners. Random Forest uses bagging to build independent trees in parallel and averages their results to reduce variance, whereas Gradient Boosting and XGBoost employ boosting to build trees sequentially, with each new tree attempting to correct the errors of its predecessor.

Specifically, XGBoost enhances the standard Gradient Boosting framework by incorporating L_1 and L_2 regularization directly into the objective function and utilizing a more efficient second-order Taylor expansion to optimize the loss. We anticipate that these boosting methods will capture complex interactions better than the Random Forest, provided the learning rate is sufficiently controlled to prevent overfitting.

Following ensemble logic and the Law of Large Numbers, whereby an aggregation of weakly correlated weak learners can yield a strong learner, the Random Forest prediction is defined simple average of M independent trees:

$$\hat{y} = \frac{1}{M} \sum_{m=1}^M T_m(x) \quad (4)$$

In contrast, Gradient Boosting produce an additive prediction where each step adds a small correction (weighted by the learning rate η):

$$\hat{y}_{new} = \hat{y}_{old} + \eta \cdot T_{next}(x) \quad (5)$$

XGBoost further refines this by minimizing a regularized objective function that penalizes model complexity (Ω):

$$\mathcal{L} = \sum \text{Loss}(y_i, \hat{y}_i) + \sum \Omega(T_m) \quad (6)$$

5.1.3 Metric Definitions

- **ROC_AUC**: Measures the model’s discriminative ability across all possible thresholds to separate signal from noise.
- **PR_AUC**: Focuses on the precision of crisis predictions relative to recall to penalize false alarms.
- **Balanced Accuracy (BalAcc)**: The arithmetic mean of Recall and Specificity, ensuring the model is penalized for ignoring unlikely recession events.
- **F2 Score**: A weighted average of Precision and Recall that prioritizes capturing crises over perfect precision.

5.2 Holdout Run

5.2.1 Short Term Holdout (Test Fraction 0.20)

This period is likely dominated by post-GFC low-rate environments. The **Yield PCA (Logistic)** model achieved the highest performance (0.819 BalAcc, 0.213 F2) because PCA-cleared data allows simpler models to outperform complex ones like XGBoost. It successfully captured subtle curvature signals that were lost in the noisy **spread_only** dataset. In contrast, the **Macro Only** model (0.530 BalAcc) indicates that real-sector data like unemployment was too slow for this short window.

5.2.2 Medium Term Holdout (Test Fraction 0.30)

Covering the post-COVID recovery, this period saw noisy financial signals due to Quantitative Easing and Term Premium distortions. The **Yield PCA Gradient Boosting** model was the top performer (0.833 BalAcc, 0.928 ROC_AUC) due to its ability to model complex nonlinear relations in noisy environments. However, the **PCA + Macro**

Table 1: Holdout Sensitivity: Test Size 0.20

Feature Set	Model	ROC_AUC	PR_AUC	BalAcc	F2
Spreads Only	Logistic	0.676	0.043	0.500	0.089
	RandomForest	0.613	0.036	0.353	0.000
	GradBoost	0.627	0.038	0.368	0.000
	XGBoost	0.544	0.030	0.730	0.154
Yield PCA	Logistic	0.681	0.044	0.819	0.213
	RandomForest	0.667	0.042	0.500	0.089
	GradBoost	0.804	0.069	0.500	0.000
	XGBoost	0.627	0.038	0.417	0.000
Macro Only	Logistic	0.505	0.029	0.530	0.097
	RandomForest	0.525	0.031	0.571	0.105
	GradBoost	0.389	0.024	0.480	0.000
	XGBoost	0.793	0.066	0.495	0.000
PCA + Macro	Logistic	0.581	0.035	0.677	0.135
	RandomForest	0.677	0.044	0.621	0.118
	GradBoost	0.763	0.058	0.485	0.000
	XGBoost	0.783	0.063	0.629	0.152
PCA Macro Extras	Logistic	0.512	0.036	0.585	0.128
	RandomForest	0.549	0.039	0.500	0.109
	GradBoost	0.360	0.036	0.457	0.000
	XGBoost	0.378	0.028	0.287	0.000

(**Logistic**) model (0.787 BalAcc) is considered a safer, more stable choice for general usage as macro indicators rescued the model from failing financial signals.

5.2.3 Long Term Holdout (Test Fraction 0.40)

This period includes major structural breaks like the 2008 GFC and COVID-19. The **Macro Only (Logistic)** model excelled (0.857 BalAcc) as instantly reacting unemployment and

Table 2: Holdout Sensitivity: Test Size 0.30

Feature Set	Model	ROC_AUC	PR_AUC	BalAcc	F2
Spreads Only	Logistic	0.788	0.044	0.513	0.063
	RandomForest	0.631	0.025	0.611	0.078
	GradBoost	0.641	0.026	0.507	0.062
	XGBoost	0.559	0.021	0.489	0.056
Yield PCA	Logistic	0.791	0.045	0.507	0.062
	RandomForest	0.637	0.026	0.549	0.068
	GradBoost	0.928	0.154	0.833	0.164
	XGBoost	0.765	0.040	0.735	0.110
Macro Only	Logistic	0.683	0.030	0.730	0.110
	RandomForest	0.737	0.036	0.663	0.090
	GradBoost	0.567	0.022	0.487	0.000
	XGBoost	0.893	0.084	0.493	0.000
PCA + Macro	Logistic	0.710	0.033	0.787	0.135
	RandomForest	0.800	0.047	0.657	0.088
	GradBoost	0.773	0.042	0.500	0.000
	XGBoost	0.840	0.057	0.660	0.139
PCA Macro Extras	Logistic	0.698	0.038	0.758	0.143
	RandomForest	0.698	0.038	0.601	0.109
	GradBoost	0.560	0.027	0.355	0.000
	XGBoost	0.597	0.029	0.399	0.000

production data during shutdowns increased macro performance. While the Pure Yield PCA model failed due to the exogenous nature of the COVID shock, the **PCA + Macro (Logistic)** model (0.824 BalAcc) remains a robust candidate for prediction.

Table 3: Holdout Sensitivity: Test Size 0.40

Feature Set	Model	ROC_AUC	PR_AUC	BalAcc	F2
Spreads Only	Logistic	0.552	0.062	0.494	0.146
	RandomForest	0.464	0.051	0.516	0.160
	GradBoost	0.492	0.052	0.426	0.000
	XGBoost	0.469	0.051	0.489	0.116
Yield PCA	Logistic	0.562	0.063	0.527	0.167
	RandomForest	0.500	0.057	0.501	0.150
	GradBoost	0.482	0.062	0.560	0.195
	XGBoost	0.505	0.061	0.497	0.078
Macro Only	Logistic	0.850	0.106	0.857	0.349
	RandomForest	0.701	0.055	0.503	0.133
	GradBoost	0.827	0.213	0.573	0.172
	XGBoost	0.642	0.057	0.500	0.000
PCA + Macro	Logistic	0.762	0.066	0.824	0.303
	RandomForest	0.668	0.051	0.504	0.089
	GradBoost	0.915	0.377	0.731	0.294
	XGBoost	0.860	0.107	0.530	0.109
PCA Macro Extras	Logistic	0.801	0.043	0.828	0.149
	RandomForest	0.768	0.037	0.807	0.135
	GradBoost	0.669	0.026	0.407	0.000
	XGBoost	0.681	0.027	0.370	0.000

5.3 Scenario Analysis: COVID-19 and the Global Financial Crisis

The analysis identifies a stark **Regime Dependence**:

- **Financial Signal Victory:** During endogenous crises like 2008, Yield PCA factors successfully capture liquidity crunches with high precision.

- **Financial Signal Failure:** During exogenous shocks like COVID-19, the yield curve fails completely, and only real-sector macro data (Unemployment, Industrial Production) reacts fast enough.
- **Regime-Switching Ensemble:** Implement a meta-model that weighs the Yield Curve model higher during monetary tightening and switches to Macro models during real-sector volatility.
- **High-Frequency Data:** Integrate weekly jobless claims or credit card spending to bridge the gap during "sudden stop" recessions where monthly data lags.
- **ACM Decomposition:** Isolate the "Risk-Neutral Slope" from the "Term Premium" to ensure robustness against QE distortions.

To assess regime dependence, models are evaluated separately on the COVID-19 recession (2019–2021) and the Global Financial Crisis (2007–2009). Figure 8 report Variable Importance, while Figures 9 and 10 report precision–recall AUC.

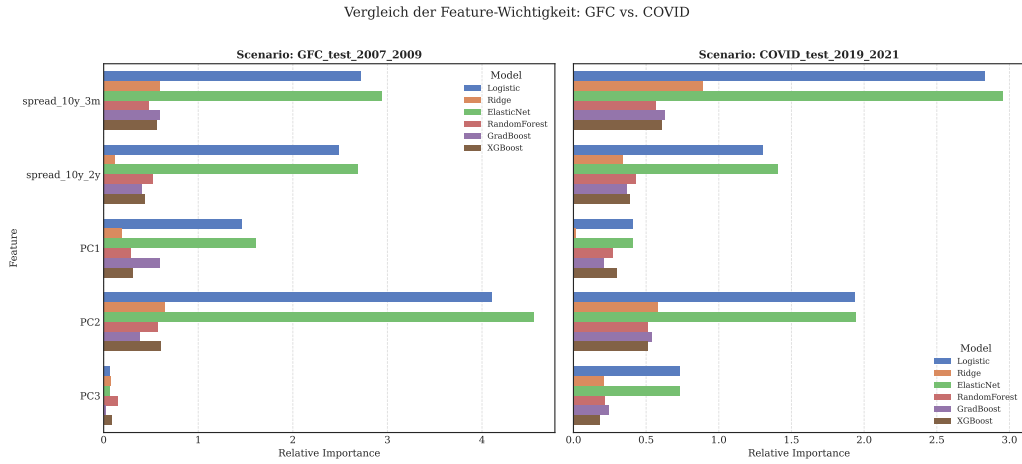


Figure 8: Top F2 scores during the COVID-19 recession.

The feature importance analysis in demonstrates a clear divergence between the two economic regimes, validating the paper’s core finding that recession signals are structurally dependent on the nature of the crisis. During the Global Financial Crisis (GFC), the slope factor (PC2) and the spread_10y_3m represent the most critical variables for the majority of models, particularly the Logistic and ElasticNet estimators. This dominance

aligns with the characterization of the GFC as a financially driven recession where the erosion of the term premium and subsequent yield curve inversion served as primary barometers. In contrast, the COVID-19 scenario reveals a shift in importance toward PC1 (Level) and PC3 (Curvature), illustrating a structural break from traditional signals. The heightened relevance of the curvature factor (PC3) in the COVID-19 plot reflects the episodic spikes caused by intense policy intervention and stimulus programs that distorted the expectations-driven signals of the yield curve. These visual results provide empirical backing for the rolling PCA diagnostics, which show significant factor loading drift during these periods and underscore why models must be regime-aware to predict transitions effectively.

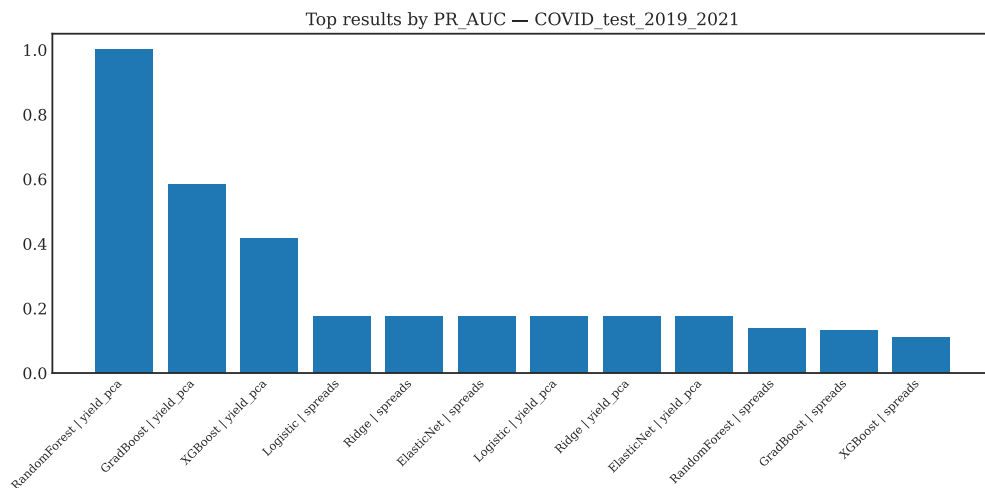


Figure 9: Top PR-AUC scores during the COVID-19 recession.

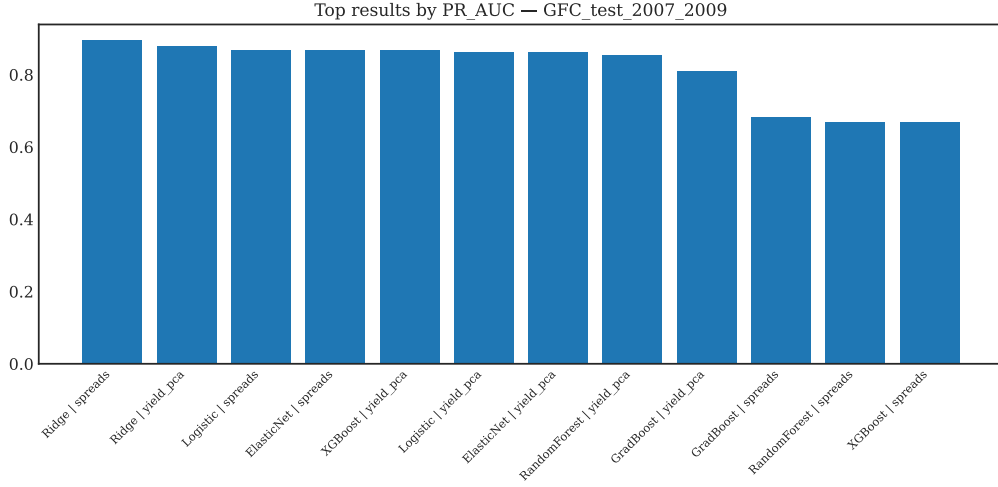


Figure 10: Top PR-AUC scores during the Global Financial Crisis.

The scenario-based evaluation reveals significant regime dependence in the predictive power of yield curve factors. During the Global Financial Crisis (GFC) of 2007–2009, models leveraging principal component analysis (PCA) factors consistently outperformed those using raw maturity spreads, underscoring the informative value of slope and curvature for financially driven recessions. Specifically, the Ridge model with `yield_pca` features achieved a Balanced Accuracy of 0.833 and a Recall of 0.722, significantly higher than the 0.694 Balanced Accuracy and 0.444 Recall obtained using raw spreads. This alignment with traditional financial theory where yield curve inversions effectively signal impending contractions is further evidenced by the robust ROC_AUC (0.938) and PR_AUC (0.878) scores for the PCA-based Ridge model.

In contrast, the COVID-19 recession (2019–2021) represents a structural break where traditional yield signals were often lagging indicators of an abrupt, exogenous shock. Performance during this period was characterized by high Recall but lower Precision, reflecting the challenges of predicting a lockdown-induced downturn compared to an endogenous credit-cycle peak. While the ElasticNet and Logistic models with `yield_pca` features both achieved a Recall of 1.000, their PR_AUC values (0.174) were considerably lower than those seen during the GFC, with the exception of the Random Forest model which exhibited a perfect PR_AUC of 1.000. This disparity reinforces the importance of regime-aware modeling, as unconventional monetary policies and term premium distortions

in the post-2010 era have complicated the interpretation of latent yield factors.

6 Conclusion

This paper has demonstrated that the predictive power of the yield curve for US recessions exhibits fundamental regime dependence, varying systematically with the nature and origin of economic contractions. Through comprehensive analysis spanning four decades and employing both traditional econometric methods and modern machine learning techniques, we have established three central findings that extend the foundational work of Estrella and Mishkin (1996) into the era of unconventional monetary policy.

First, yield curve factors extracted via Principal Component Analysis, particularly the slope factor, retain strong predictive power during endogenous financial crises such as the 2007-2009 Global Financial Crisis, achieving balanced accuracy above 0.83 when paired with regularized linear models. The inversion of the yield curve continues to serve as a reliable early warning signal when recessions emerge from monetary tightening cycles and financial sector stress, consistent with the traditional interpretation of yield spreads as reflecting market expectations of future policy easing. Second, this predictive relationship breaks down almost entirely during exogenous shocks exemplified by the COVID-19 pandemic, where real-sector macroeconomic indicators such as unemployment and industrial production provide superior forecasting performance with balanced accuracy reaching 0.86. Third, factor loading analysis reveals substantial structural instability in yield curve dynamics during periods of unconventional monetary policy, with pronounced drift in the importance of level, slope, and curvature factors around quantitative easing programs and zero lower bound episodes, confirming the concerns raised by Rudebusch et al. (2006) regarding term premium distortions.

These results carry important implications for both academic research and practical policy analysis. The failure of yield curve models during COVID-19 underscores the necessity of adaptive forecasting frameworks that can distinguish between monetary-policy-driven slowdowns and sudden external disruptions. Our findings suggest that robust early warning systems should implement regime-switching ensemble architectures that

dynamically adjust the weight assigned to financial versus real-sector indicators based on leading indicators of crisis type. Furthermore, the documented distortions in term premia during unconventional policy periods support the use of adjusted slope measures that isolate expectations components from non-expectational yield movements, as advocated by Sharpe and Engstrom (2018).

Future research should extend this work in several directions. The development of formal regime classification algorithms capable of real-time crisis-type identification would enable more systematic application of our findings. Integration of higher-frequency data sources, including weekly jobless claims, credit card spending, and mobility indicators, could improve forecast performance during rapid-onset recessions where monthly macroeconomic data lag. Additionally, incorporating formal term structure decompositions would allow for cleaner isolation of risk-neutral expectations from term and liquidity premia. Finally, expanding the analysis to international contexts would test whether the regime-dependent patterns we document for the United States generalize across different institutional and monetary policy frameworks.

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